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Cancer Survivorship at Stanford Cancer Institute

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Abstract

The Stanford Cancer Survivorship Program is a key initiative of Stanford Cancer Institute. The program's mission is to improve the experience and outcomes of patients and family caregivers throughout all phases of the cancer trajectory by advancing survivorship research, clinical care, and education. The four pillars of the program include clinical care delivery with a focus on primary care–survivorship collaboration and expanding specialty services, education and training of healthcare professionals, transdisciplinary patient-oriented research, and community engagement. Cross-cutting areas of expertise include: a) adolescents and young adults (AYAs), b) mental health and patient self-management, c) integration of primary care, and d) postgraduate

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medical education. The clinical care model includes embedded survivorship clinics within disease groups in outpatient clinics, novel clinics designed to address unmet needs such as sexual health for women, and primary care-based faculty-led survivorship clinics for patients undergoing active cancer care requiring co-management, those who have completed active therapy and those at high risk for cancer due to genetic risk. Educational initiatives developed to date include an online course and medical textbook for primary care clinicians, a lecture series, monthly research team meetings, and rotations for medical trainees. Patient-facing activities include webinars and a podcast series designed to promote awareness, thus expanding the provision of expert-vetted information. Ongoing research focuses on oncofertility and family building after cancer, improving communication for AYAs, changing mindsets to improve quality of life through targeted digital interventions, increasing capacity to care for cancer survivors, and strengthening collaboration with community partners.

Introduction

The Stanford Cancer Survivorship Program is a key initiative of Stanford Cancer Institute (SCI), an NCI-designated Comprehensive Cancer Center in Northern California. The SCI catchment area encompasses 10 counties and 7.8 million people. Area-level demographics vary widely across the region and include urban, suburban, and rural settings. An estimated 72.8% of patients seen at SCI identify as white (includes white/Hispanic), 18.9% Asian, 14.1% Hispanic/Latino, 3.1% Black/African American, 1.8% multiracial/other, and 0.8% Native Hawaiian/Pacific Islander. The main cancer center is in Palo Alto, with network sites in San Jose, Redwood City, Emeryville, Pleasanton, Castro Valley, and Turlock, California.

Responding to the call to improve care coordination for cancer survivors and advance the national research agenda after the 2006 National Academies Press report “Lost in Transition” [1], Stanford launched disease-specific survivorship clinics in 2012 and later a general survivorship clinic, and a research program nested within the Population Science program. These were consolidated as one program with the recruitment of a new director in the Fall of 2016.

Tasked with improving health outcomes for all people treated for cancer and advancing research, the program includes professionals from various disciplines and involves collaborations with scientists across Stanford University. Key initiatives include patient-facing services such as the expansion of clinical services including primary care, establishment of new transdisciplinary research teams, and bidirectional partnerships with community organizations that inform the program’s activities and research platform.

With funding from Stanford Medicine, the parent organization that includes Stanford Health Care, Stanford Children’s Health, and Stanford University School of Medicine and its academic departments and divisions, the program is structured around four specific aims or pillars: clinical care delivery, transdisciplinary patient-oriented research, education and training of professionals, and community engagement (Figure 1). Specific areas of expertise that cross each of these pillars and represent the research interests of Stanford faculty include adolescents and young adults (AYAs), mental health and patient self-management, integration of primary care clinicians in survivorship care, and capacity building through

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educating healthcare professionals to provide care that meets recommended standards of practice and is culturally attuned. A hallmark of the program is collaboration across disciplines and departments at Stanford Medicine and throughout Stanford University.

Clinical Care Delivery

1. Disease-specific, cancer center-based survivorship clinics

The clinical program has evolved in compliance with the Committee on Cancer's requirements and conforms to the latest version updated in 2023[2]. The Stanford Cancer Survivorship Program has a designated director who is a leader at the Cancer Center and represents Stanford in the NCCN Survivorship Guideline panel. The director works with an operations team tasked with implementing practices and tracking process metrics. Obtaining accurate volume data has proven to be challenging after shifting visits from a general designated clinic to disease-specific clinics. Although we have created a dashboard to capture outpatient survivorship visits, we routinely miss visits coded with disease-specific billing codes. Since 2017, we do not routinely capture data for completion of Survivorship Care Plans, although we recommend these be completed in the electronic health record and shared with patients and their primary care physicians.

Starting in 2012, disease-specific clinics provided survivorship care within the breast, gynecologic, hematology, lymphoma, gastrointestinal, and blood and marrow transplant (BMT) clinics. The visits were conducted by advanced practice providers (APPs). While patient surveys showed high satisfaction with the visits and increased knowledge about past treatment and future health recommendations, referrals from oncologists were limited. In 2016, the disease-specific clinics were consolidated into a general survivorship clinic with a peak visit volume of 710 that year and a gradual attrition of patients in 2017-2018 that led to sunsetting the general survivorship clinic in 2019 and bolstering efforts to train APPs within disease groups. The rate of adoption of the embedded APP and physician model varies among disease groups and is generally increasing, with successful re-integration in breast, genitourinary, BMT, lymphoma, and gynecologic disease groups at the flagship cancer center and network sites. Identifying champions within disease groups was instrumental in gaining acceptance from oncologists and implementing a seamless model that works efficiently for clinical care programs. Survivorship visits are conducted by APPs and physicians, with referral pathways integrated into the electronic health record. Available volume estimates for the past year show 270 visits in breast oncology, 258 in genitourinary oncology, 147 in BMT, and 29 in lymphoma. Survivorship is one of several supportive care services at the Stanford Comprehensive Cancer Center. Other specialty clinics that complement and enhance survivorship care through referrals and co-management include psycho-oncology, cardio-oncology, nephro-oncology, cutaneous oncology, reproductive medicine, sexual health, cancer genetics, and the Stanford Adolescent and Young Adult Cancer (SAYAC) Program.

2. Primary care-based survivorship clinics

A collaboration between the divisions of oncology and primary care within the Department of Medicine led to the establishment of a faculty practice dedicated to primary care for

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cancer survivors in 2018. Visits are designed to 1) ease transition from oncology-led care back to primary care-led care, 2) address the impact of cancer and cancer treatment on physical and mental health and deliver preventive care, 3) consider cancer survivorship through the lens of chronic disease management, 4) foster patient empowerment and self-management, and 5) facilitate coordination and communication between primary care and subspecialists.

The three primary care physicians (PCPs) in the practice have received training in survivorship, and one is board-certified in oncology and hematology. Initially, the PCPs accepted patients into their primary care panels to meet ongoing primary care and survivorship needs in one place. They subsequently transitioned to a consultative model of care due to increased demand, and to date have seen over 500 new patients. The consultative model includes one to several visits (individualized for the patient) to discuss survivorship topics and create a survivorship care summary and recommendations that a patient can share with their PCP; this approach is novel in being developed by PCPs for PCPs. Patients include individuals at risk for cancer due to genetic susceptibility, those with chronic medical problems requiring co-management during active cancer treatment, and long-term survivors many years after completion of treatment, including adult survivors of childhood cancer. Sources of referrals include oncologists, hematologists, and BMT teams from the cancer center, high-risk clinics, and the AYA program whose clinicians appreciate a warm hand-off to PCPs with experience in treating this population.

3. Stanford Adolescent and Young Adult Cancer (SAYAC) Program

Stanford's pediatric survivorship clinic, "Health After Therapy" or HAT, is a dedicated clinic within the Bass Center for Childhood Cancer and Blood Diseases. Childhood cancer survivors of all disease types are typically referred at 5 years off-therapy and are seen by a pediatric oncology APP annually or biennially depending on risk of late effects.

Teenagers and young adults are seen by an APP who leads SAYAC, a joint program between Lucile Packard Children's Hospital Stanford and Stanford Health Care to help meet the unique needs of patients ages 15-30, with an estimated volume of 500 visits per year. Visits focus on communication and interventions related to fertility and sexual health, peer to peer support, psychosocial needs, school or work issues, and referrals to specialties as needed, such as mental health and reproductive endocrinology.

Recognizing a need to diminish attrition in survivorship-based care for young adults, SAYAC launched a pilot clinic ("Getting Regular Evaluations After Treatment" or GREAT clinic) to facilitate young adult childhood cancer survivors' transitions from pediatric- to adult-focused survivorship care and to improve care coordination with primary care [3]. This clinical pilot was supported by a Teen Cancer America program grant, and for logistical reasons, focused on AYA survivors of hematologic malignancies and BMT. Patients were seen within the hematology or BMT clinics of the two adult oncology champions for the clinic, with a pediatric oncologist leading the medical survivorship visit and discussing patients with the adult hematology/BMT physician, and an AYA APP leading a back-to-back visit focused on AYA concerns. A key component of the clinic was to facilitate bidirectional communication with primary care to improve care coordination. While the pilot period

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for this clinic has concluded, the Stanford Cancer Survivorship Program and SAYAC have identified a joint AYA survivorship clinic as a promising opportunity for future program development.

Transdisciplinary Research

Survivorship research is integrated in the Population Science Program at SCI. The team holds monthly research meetings that bring together investigators from diverse backgrounds to share research concepts and works-in-progress, and to meet with invited guests interested in collaboration. SCI sponsors “Innovation Awards” biannually to fund promising pilot projects on cancer-related topics. Five such awards supported cancer survivorship investigators and catalyzed collaborations, and preliminary data enabled investigators to apply for 2 NIH R- and 1 K-level funding awards. While the investigators have a variety of medical and scientific backgrounds, a common theme across the survivorship-focused research at SCI is that it is grounded in patient and family needs and focuses on clinical interventions and health equity [4,5]. Examples of successful research collaborations that received, or are currently under consideration for external funding include projects related to: a) oncofertility experiences and family building after cancer [6], b) addressing unmet needs of cancer survivors with breast and gynecologic cancers [7–10], c) improving survivorship communication for Hispanic/Latino AYAs in the Salinas Valley in partnership with a community-based organization, d) understanding the experiences of South Asian women with breast cancer and their family caregivers, and e) cancer mindsets: a novel digital intervention for cancer survivors [10,11]. Since 2017, the core members of the program (CB, LCH, OP, LS, SMS) published 55 first or senior author manuscripts related to survivorship and an additional 146 manuscripts where they were not lead or senior author, after removing duplicates.

In partnership with major cancer centers, Stanford will host the second biennial Cancer Center Survivorship Research Forum in September 2023, a 2-day meeting focused on implementing equitable cancer survivorship care.

Education and Training

The Stanford Cancer Survivorship Program has designed educational programs to increase awareness and improve skills and competence among healthcare professionals. These include: 1) a monthly continuing medical education (CME)-accredited cancer survivorship speaker series now integrated into the oncology division conference series to reach trainees, 2) an online CME course (“Health After Cancer: Cancer Survivorship for Primary Care”) that reached over 1100 participants [12,13], 3) a textbook written for a generalist audience (“Essentials of Cancer Survivorship”) [14], 4) workshops to hone skills for nurses and APPs, and 5) invited lectures for local and national audiences. The primary care-based survivorship clinicians deliver didactic talks for Internal Medicine residents and medical students and provide opportunities for trainees to rotate in the clinic.

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Community Engagement and Patient Self-management

Engaging the cancer survivor community to inform care delivery and set research priorities is foundational for the program. Stanford's Patient Family Advisory Council advises the program on issues related to clinical care and patient/family education. Other collaborations include SCI's Community Advisory Board, Bay Area Cancer Connections (https://www.bayareacancer.org/), Silicon Valley Survivorship Coalition, and Jacob's Heart Children's Cancer Support Services, a community-based organization that supports pediatric and AYA patients and families (https://www.jacobsheart.org/). On a national level, team members have participated in programs offered by advocacy groups including Ovarcome (https://ovarcome.org/), Living Beyond Breast Cancer (https://www.lbbc.org/), Stupid Cancer (https://stupidcancer.org/), Cancer Care (https://www.cancercare.org/, Sharsheret, the Jewish Breast and Ovarian Cancer Community (https://sharsheret.org/), Share (https://www.sharecancersupport.org/), Cactus Cancer Society (https://www.cactuscancer.org), Cancer Support Community (https://www.cancersupportcommunity.org/), and with professional societies including the American Society of Clinical Oncology, the Oncology Nursing Society, and Cancer Survivorship Provider Network. Outreach educational programs include webinars on mental wellbeing for cancer patients, a podcast series for survivors, disease-specific symposia for survivors of head & neck cancers, prostate cancer, and a yearly celebration for bone marrow transplant survivors. SCI has sponsored an annual Breast Cancer for African Americans program for the past 12 years.

Lessons and Future Directions

Over the past seven years, the Stanford Cancer Survivorship Program has grown considerably and continues to evolve to meet the needs of the cancer center and its expanding network. The research teams and infrastructure are now maturing to the point where investigators are successfully competing for external grant funding. Still, there is a lot more we aim to do.

The COVID-19 pandemic slowed our growth by diverting attention to solving urgent matters related to provision of services for patients undergoing active cancer treatment, but also provided new opportunities for us to collaborate on a national level on advancing research and scholarship. The emergence of telemedicine during the pandemic has been a boon for all survivorship clinicians, as this medium is well suited for patient encounters that do not require physical exams or laboratory testing.

In reflecting on our program, one of our key learnings has been the need to remain flexible in design of clinical care delivery models and to negotiate with the administration in choosing metrics that reflect survivorship 'activity' across disease groups and practices. This is still a work in progress and requires a fluid relationship between clinicians and administrators and considerable time and patience. As a program, we have prioritized trans-disciplinary collaboration in survivorship research and education, which has sparked innovative projects and contributed to training and educating the next generation of cancer survivorship clinicians and researchers.

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As we look to the future, we are committed to building an integrated clinical and research program to support the needs of AYA cancer survivors through emerging partnerships between the adult and pediatric programs. Recognizing the widespread gaps in provision of mental healthcare for survivors across the lifespan, we hope to integrate mental healthcare and psychosocial support into routine clinical settings through innovative research and technology. While we have expanded clinical psychosocial services, access remains limited by resources and insurance coverage – barriers that must be addressed to achieve equity in cancer care. One of our areas of research is in digital interventions targeting cancer mindsets at the point of diagnosis[11]. We are interested in collaborating with others to discover, test, and implement new interventions that are scalable, sustainable, affordable, and effective. One such avenue is through cultivating community-academic partnerships to advance shared research agendas tailored to local needs. We are also actively seeking ways to improve sustainability and reach of our primary care-based survivorship program through collaborations with community partners. Additional specific initiatives that are ongoing include: a) recruitment of an academic leader for the pediatric survivorship program, b) expansion of clinical services to disease groups that currently do not offer such services, c) creation of a dashboard that captures survivorship activities using metrics other than completion of treatment summaries and survivorship care plans, and d) establishing a central free consultative service for primary care community clinicians seeking advice regarding patients who are cancer survivors.

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Figure 1.
Stanford Cancer Survivorship Program Elements

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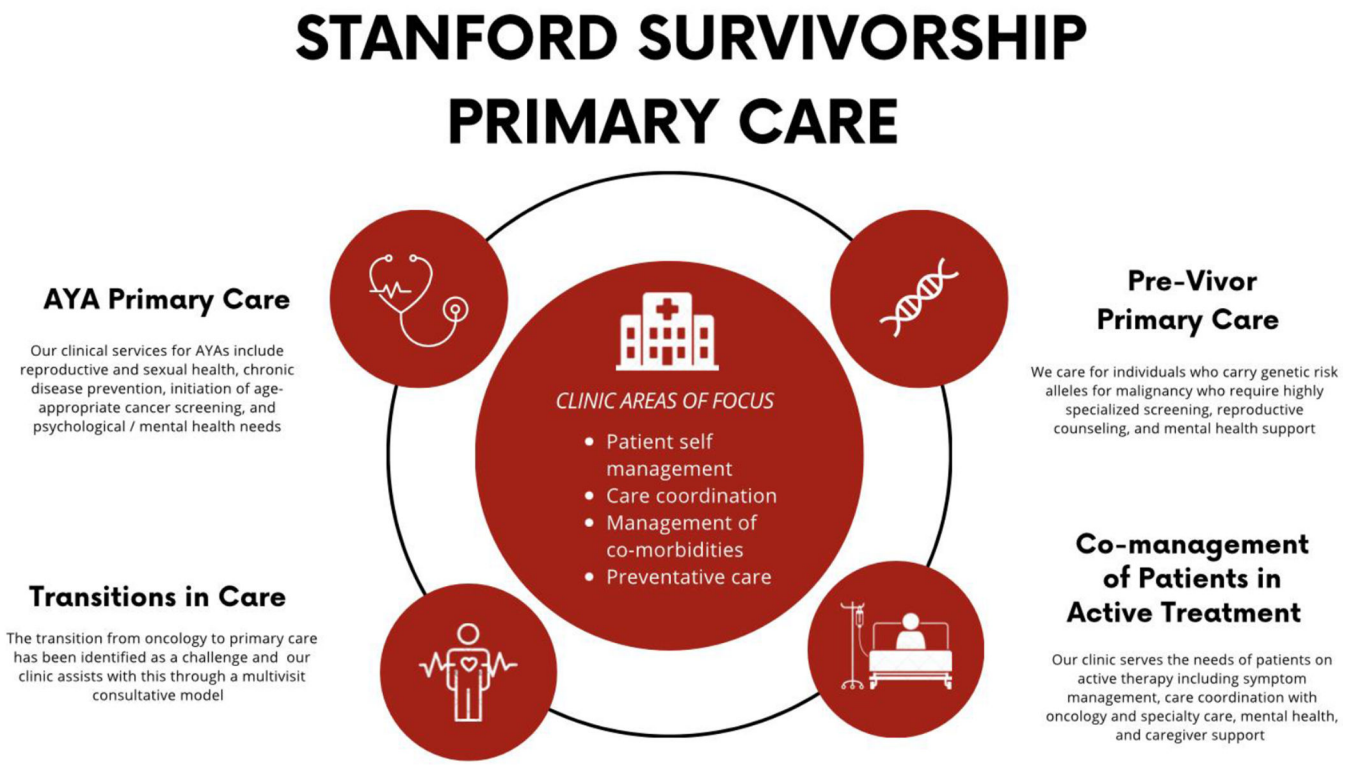

Figure 2.
Primary Care-Based Survivorship Clinics at Stanford